

Almost everywhere monotone additive functions

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Abstract

We prove that a real additive function which decreases on a set of natural density zero is a nonnegative constant multiple of the logarithm. This settles a conjecture of Erdős from 1946, listed as Problem 1122 on the Erdős Problems website. The main step is to deduce finite concentration without a growth assumption on the values at higher prime powers. A truncated normalization and a bounded clipping reduce this step to Mangerel’s first-moment theorem for short intervals, Ruzsa’s second-moment estimate, and Elliott’s fourth-moment inequality.

1 Introduction

Throughout, $\mathbb{N} = \{1, 2, \dots\}$. A function $f: \mathbb{N} \rightarrow \mathbb{R}$ is *additive* if $f(ab) = f(a) + f(b)$ whenever $(a, b) = 1$. Erdős proved that an additive function which is nondecreasing on $\mathbb{N}$ must be a constant multiple of $\log n$. He conjectured that the same conclusion holds when the set of decreases has natural density zero [1, p. 3]. The conjecture is also recorded as Erdős Problem 1122 [10].

Theorem 1.1. *Let $f: \mathbb{N} \rightarrow \mathbb{R}$ be additive. If*

$$D_f(X) := \#\{n \leq X : f(n+1) < f(n)\} = o(X),$$

then there is a constant $c \geq 0$ such that $f(n) = c \log n$ for every $n \in \mathbb{N}$.

A Lean 4 formalization of the argument, treating the cited results of Mangerel, Ruzsa, Erdős, and Hildebrand as hypotheses, is available online.¹

Erdős proved that $f(n+1) - f(n) = o(1)$ forces $f = c \log$. Kátai obtained the same conclusion from $X^{-1} \sum_{n \leq X} |f(n+1) - f(n)| \rightarrow 0$ [5]. Wirsing gave another proof [9]; see also Elliott [3, Ch. 11, pp. 244–247]. Hildebrand proved the density-one version [4]. Mangerel obtained an approximate logarithmic representation under the present density hypothesis [6, Theorem 1.9], and the exact conclusion for completely additive functions under additional conditions on prime values and decreases [6, Corollary 1.7].

We say that f has *finite concentration* if there are constants $d > 0$ and $R < \infty$ such that, for arbitrarily large X , some interval of length R contains at least dX of the values $f(n)$, $n \leq X$. The interval may depend on X . Erdős’s Theorem V [1] makes finite concentration equivalent to the existence of $c \in \mathbb{R}$ such that

$$\sum_p \frac{\min\{(f(p) - c \log p)^2, 1\}}{p} < \infty. \tag{1.1}$$

¹<https://github.com/FireflySentinel/erdos-1122>.

Under (1.1), Hildebrand gives a symmetric limiting law for $f(n+1) - f(n)$; the density hypothesis and his corollary then force $f = c \log$. Thus it suffices to prove finite concentration. Assuming the contrary, we normalize f modulo a logarithmic term to fix its truncated quadratic prime mass. Ruzsa gives a positive lower bound for the variance after clipping, while Elliott supplies the fourth-moment control needed to combine Mangerel's short-interval estimate with the density hypothesis and obtain an incompatible upper bound.

Notation

Letters p, q denote primes. We write

$$\mathbb{E}_X F = \frac{1}{X} \sum_{n \leq X} F(n), \quad \mathcal{A}_H F(n) = \frac{1}{H} \sum_{j=0}^{H-1} F(n-j) \quad (n \geq H).$$

For strongly additive $z(n) = \sum_{p|n} a_p$, put $\mu_z(X) = \sum_{p \leq X} a_p/p$. All implicit constants are absolute unless a dependence is indicated. The constants K and M used in the proof are fixed before X tends to infinity. In statements about a family z_X , the prime coefficients may depend on X .

2 Moment estimates and short intervals

For an additive function z , define

$$A_0(z, X) = \sum_{p^k \leq X} \frac{z(p^k)}{p^k} \left(1 - \frac{1}{p}\right), \quad B_2(z, X)^2 = \sum_{p^k \leq X} \frac{|z(p^k)|^2}{p^k}.$$

Ruzsa's second-moment estimate [7], in the form recorded by Mangerel [6, Lemma 3.3], gives, for all sufficiently large X and uniformly in z ,

$$\mathbb{E}_X |z - A_0(z, X)|^2 \asymp \inf_{b \in \mathbb{R}} \left(b^2 + \sum_{p^k \leq X} \frac{|z(p^k) - b \log(p^k)|^2}{p^k} \right). \quad (2.1)$$

The upper bound with $b = 0$ is the Turán–Kubilius inequality, valid for all $X \geq 2$ [8, Ch. III.3]. For a strongly additive function $z(n) = \sum_{p|n} a_p$, summing the geometric series gives

$$A_0(z, X) - \mu_z(X) = - \sum_{p \leq X} a_p p^{-k_p}, \quad k_p = \min\{k \geq 1 : p^k > X\}.$$

Thus $|A_0(z, X) - \mu_z(X)| \leq X^{-1} \sum_{p \leq X} |a_p| \leq (\sum_{p \leq X} a_p^2/p)^{1/2}$ by Cauchy–Schwarz and $\sum_{p \leq X} p \leq X^2$. When $|a_p| \leq L$, the same difference is at most $L\pi(X)/X = o(1)$. Elliott's unweighted center differs from $\mu_z(X)$ by at most $\sum_{p \leq X} |a_p|/[p(p-1)] \ll (\sum_{p \leq X} a_p^2/p)^{1/2}$, so his fourth-moment inequality [2, Theorem 1] gives

$$\mathbb{E}_X |z - \mu_z(X)|^4 \ll \left(\sum_{p \leq X} \frac{a_p^2}{p} \right)^2 + \sum_{p \leq X} \frac{a_p^4}{p}, \quad X \geq 2. \quad (2.2)$$

Also $B_2(z, X)^2 \leq 2 \sum_{p \leq X} a_p^2/p$. Combining Ruzsa's upper bound with (2.2), if $|a_p| \leq L$ and $\sum_{p \leq X} a_p^2/p \leq V$, then

$$\mathbb{E}_X |z - \mu_z(X)|^2 \ll V, \quad \mathbb{E}_X |z - \mu_z(X)|^4 \ll V^2 + L^2 V. \quad (2.3)$$

Lemma 2.1. *Let z_X be strongly additive functions satisfying*

$$|z_X(p)| \leq L, \quad \sum_{p \leq X} \frac{z_X(p)^2}{p} \leq V$$

for fixed $L, V < \infty$. Then

$$\lim_{H \rightarrow \infty} \limsup_{X \rightarrow \infty} \frac{1}{X} \sum_{H \leq n \leq X} |\mathcal{A}_H z_X(n) - \mu_{z_X}(X)|^2 = 0, \quad (2.4)$$

where H runs through the positive integers.

Proof. For $Y \leq X$, write

$$d_Y = \frac{2}{Y} \sum_{Y/2 < n \leq Y} z_X(n), \quad F_Y(n) = \mathcal{A}_H z_X(n) - d_Y.$$

Mangerel's Theorem 1.1 [6] gives, for every integer $10 \leq H \leq Y/100$,

$$\frac{2}{Y} \sum_{Y/2 < n \leq Y} |F_Y(n)| \ll \left(\sqrt{\frac{\log \log H}{\log H}} + (\log Y)^{-1/800} \right) \sqrt{2V}. \quad (2.5)$$

Here $B_2(z_X, Y)^2 \leq 2V$, and the estimate is uniform over additive functions, so it applies to the family z_X .

Counting multiples of each prime gives

$$d_Y - \mu_{z_X}(Y) = \sum_{p \leq Y} z_X(p) \left(\frac{2}{Y} \left(\left\lfloor \frac{Y}{p} \right\rfloor - \left\lfloor \frac{Y}{2p} \right\rfloor \right) - \frac{1}{p} \right) = O\left(\frac{L\pi(Y)}{Y}\right). \quad (2.6)$$

Jensen's inequality and (2.3) at Y imply

$$\frac{2}{Y} \sum_{Y/2 < n \leq Y} |F_Y(n)|^4 \ll V^2 + L^2 V + O_L((\log Y)^{-4}). \quad (2.7)$$

Indeed, with $\mu = \mu_{z_X}(Y)$, Jensen's inequality gives

$$|\mathcal{A}_H(z_X - \mu)(n)|^4 \leq \mathcal{A}_H(|z_X - \mu|^4)(n).$$

Since $H \leq Y/100 < Y/2$, each index in these windows lies in $[1, Y]$ and occurs at most H times when we sum over $Y/2 < n \leq Y$. Equation (2.6) bounds the center shift. Hölder's inequality now gives

$$\frac{2}{Y} \sum |F_Y|^2 \leq \left(\frac{2}{Y} \sum |F_Y| \right)^{2/3} \left(\frac{2}{Y} \sum |F_Y|^4 \right)^{1/3}.$$

Together with (2.5) and (2.7), this proves the required iterated limit on each fixed dyadic band $Y \asymp X$.

To extend this conclusion to the whole interval, fix $\eta = 2^{-r}$. On each of the r dyadic bands covering $(\eta X, X]$, we have

$$|\mu_{z_X}(X) - \mu_{z_X}(Y)| \leq L \sum_{Y < p \leq X} \frac{1}{p} = o(1).$$

The omitted initial segment $H \leq n \leq \eta X$ contributes $O_{L,V}(\sqrt{\eta})$ to (2.4): apply Cauchy-Schwarz and Jensen, using the fourth-moment bound (2.3) at X . Take $X \rightarrow \infty$, then $H \rightarrow \infty$; since η is arbitrary, (2.4) follows. $\square$

3 A weighted estimate for a set of primes

Lemma 3.1. *Suppose $0 < M < 1/4$ and $\mathcal{T}$ is a set of primes at most X such that $\sum_{p \in \mathcal{T}} 1/p \leq M$. Then, uniformly in $\mathcal{T}$,*

$$\sum_{p \in \mathcal{T}} \frac{1}{p} \sum_{X/p < q \leq X} \frac{1}{q} \ll M \log(2/M) + o_{X \rightarrow \infty}(1). \quad (3.1)$$

Here M is fixed in the term $o(1)$.

Proof. We use Mertens's prime harmonic-sum formula and the standard prime-counting bound; see [8]. For $p \leq X^{1-M}$, the inner sum is at most $\log(1/M) + o(1)$. These primes contribute at most $M \log(1/M) + o(1)$.

For $p > X^{1-M}$, discard the restriction $p \in \mathcal{T}$ and interchange the sums. The terms with $q > X^M$ are bounded by

$$\left(\sum_{X^{1-M} < p \leq X} \frac{1}{p} \right) \left(\sum_{X^M < q \leq X} \frac{1}{q} \right) \ll M \log(2/M) + o(1).$$

For $q \leq X^M$, the condition $X/p < q$ implies $p > X/q$. The bound $\pi(t) \ll t/\log t$ and partial summation give, uniformly for $2 \leq q \leq X^M$,

$$\sum_{X/q < p \leq X} \frac{1}{p} \ll \frac{\log q}{\log X}.$$

Their total is $\ll (\log X)^{-1} \sum_{q \leq X^M} (\log q)/q \ll M$, proving (3.1). $\square$

4 Normalization

In the remainder of the proof, suppose that f does not have finite concentration. For $s > 0$ define

$$W_X(s) = \min_{c \in \mathbb{R}} \left\{ c^2 + \sum_{p \leq X} \frac{\min\{(f(p)/s - c \log p)^2, 1\}}{p} \right\}. \quad (4.1)$$

Lemma 4.1. *For each fixed $M > 0$ and all sufficiently large X , there are $s_X > 0$ and a minimizer c_X in (4.1) such that $W_X(s_X) = M$. Then necessarily $s_X \rightarrow \infty$, and any corresponding minimizers satisfy $c_X \rightarrow 0$. Put $u_p = f(p)/s_X - c_X \log p$ for $p \leq X$. Then*

$$\sum_{p \leq X} \frac{\min\{u_p^2, 1\}}{p} = M - c_X^2 = M - o(1), \quad (4.2)$$

$$\sum_{\substack{p \leq X \\ |u_p| < 1}} \frac{u_p \log p}{p} = c_X. \quad (4.3)$$

No u_p has absolute value exactly 1.

Proof. The expression in braces in (4.1) is continuous and coercive as a function of c , so its minimum is attained. Every minimizer satisfies $c^2 \leq \sum_{p \leq X} 1/p$, by comparison with $c = 0$. The minimum

can therefore be taken over a fixed compact interval, independent of s , and is continuous in $s > 0$. On writing $\lambda = sc$, the expression becomes

$$\frac{\lambda^2}{s^2} + \sum_{p \leq X} \frac{\min\{(f(p) - \lambda \log p)^2 / s^2, 1\}}{p}.$$

It follows that $W_X(s)$ is nonincreasing in s , and it tends to zero as $s \rightarrow \infty$ for fixed X .

For each fixed $s > 0$, $W_X(s) \rightarrow \infty$. Otherwise some unbounded sequence of X would have bounded $W_X(s)$ and bounded minimizers. Pass to a subsequence along which the minimizers converge to c . Fatou's lemma gives

$$\sum_p \frac{\min\{(f(p)/s - c \log p)^2, 1\}}{p} < \infty.$$

Rescaling by the fixed number s preserves convergence of the truncated square sum. Erdős's Theorem V would therefore give finite concentration of f , contrary to the assumption.

Continuity now supplies s_X with $W_X(s_X) = M$. Monotonicity and the preceding claim imply $s_X \rightarrow \infty$ for every such choice. We have $|c_X| \leq \sqrt{M}$. If a subsequential limit of c_X were a nonzero number c , Fatou's lemma on the fixed primes would give

$$\sum_p \frac{\min\{(c \log p)^2, 1\}}{p} \leq M,$$

which is impossible. Thus $c_X \rightarrow 0$, and (4.2) follows.

It remains to prove the stationary identity and exclude equality at the truncation threshold. Let $G(c)$ denote the expression in braces in (4.1), with $s = s_X$. For any subset J of the primes at most X , the quadratic

$$Q_J(c) = c^2 + \sum_{p \in J} \frac{(f(p)/s_X - c \log p)^2}{p} + \sum_{\substack{p \leq X \\ p \notin J}} \frac{1}{p}$$

lies above $G(c)$. Choose $J = \{p \leq X : |u_p| < 1\}$. Then $Q_J(c_X) = G(c_X)$, so c_X also minimizes Q_J . The equation $Q'_J(c_X) = 0$ is exactly (4.3). If $|u_p| = 1$ for some p , the quadratic $Q_{J \cup \{p\}}$ also touches G at c_X and must have derivative zero there, contradicting

$$Q'_{J \cup \{p\}}(c_X) - Q'_J(c_X) = -\frac{2u_p \log p}{p} \neq 0. \quad \square$$

5 Clipping the normalized function

Fix $K > 1$ and $0 < M < 1/4$, and use Lemma 4.1. Let $\phi_K(t) = \max\{-K, \min\{t, K\}\}$ and set

$$\begin{aligned} \mathcal{T} &= \{p \leq X : |u_p| > 1\}, & S_0(n) &= \sum_{\substack{p|n \\ p \notin \mathcal{T}}} u_p, \\ a_X &= \sum_{\substack{p \leq X \\ p \notin \mathcal{T}}} \frac{u_p}{p}, & S(n) &= S_0(n) - a_X. \end{aligned}$$

Coefficients at primes greater than X are set to zero. Define

$$\begin{aligned} z_X(n) &= \sum_{p|n} \phi_K(u_p), & Z(n) &= z_X(n) - a_X, \\ g_X(n) &= \phi_K \left(\frac{f(n)}{s_X} - c_X \log n - a_X \right). \end{aligned} \quad (5.1)$$

Thus g_X clips the value of the additive function, whereas z_X clips the prime contributions individually.

Lemma 5.1. *There is an absolute constant C such that*

$$\limsup_{X \rightarrow \infty} \mathbb{E}_X |g_X - Z|^2 \leq C \left(\frac{M}{K^2} + M^2 \log(2/M) + K^2 M^2 \right). \quad (5.2)$$

Proof. The small coefficients have quadratic mass at most M , and $\sum_{p \in \mathcal{T}} 1/p \leq M$. Hence

$$\mathbb{E}_X S^4 \ll M + M^2. \quad (5.3)$$

Put $\omega_{\mathcal{T}}(n) = \#\{p \in \mathcal{T} : p \mid n\}$. Counting multiples of distinct primes gives

$$\mathbb{E}_X \omega_{\mathcal{T}}(\omega_{\mathcal{T}} - 1) = \sum_{\substack{p, q \in \mathcal{T} \\ p \neq q}} \frac{1}{X} \left\lfloor \frac{X}{pq} \right\rfloor \leq M^2. \quad (5.4)$$

Moreover,

$$\mathbb{E}_X (S^2 \mathbf{1}_{\omega_{\mathcal{T}} \geq 1}) \ll M^2 \log(2/M) + o(1). \quad (5.5)$$

For $p \in \mathcal{T}$, the coefficient of p in S_0 is zero, so $S_0(pm) = S_0(m)$ for every m , including multiples of p . Writing $y = X/p$, for $y \geq 2$ equation (2.3) gives

$$\begin{aligned} \mathbb{E}_X (S^2 \mathbf{1}_{p|n}) &= \frac{1}{p} \frac{1}{y} \sum_{m \leq y} |S_0(m) - a_X|^2 \\ &\ll \frac{1}{p} \left(M + |a_X - \mu_{S_0}(y)|^2 \right). \end{aligned}$$

For $1 \leq y < 2$, the same bound follows directly from $S_0(1) = \mu_{S_0}(y) = 0$. By Cauchy–Schwarz,

$$|a_X - \mu_{S_0}(X/p)|^2 \leq M \sum_{X/p < q \leq X} \frac{1}{q}.$$

Hence, using $\mathbf{1}_{\omega_{\mathcal{T}} \geq 1} \leq \sum_{p \in \mathcal{T}} \mathbf{1}_{p|n}$,

$$\begin{aligned} \mathbb{E}_X (S^2 \mathbf{1}_{\omega_{\mathcal{T}} \geq 1}) &\ll M \sum_{p \in \mathcal{T}} \frac{1}{p} + M \sum_{p \in \mathcal{T}} \frac{1}{p} \sum_{X/p < q \leq X} \frac{1}{q} \\ &\ll M^2 + M^2 \log(2/M) + o(1), \end{aligned}$$

by Lemma 3.1. This proves (5.5).

Set

$$g_X^*(n) = \phi_K \left(S(n) + \sum_{\substack{p|n \\ p \in \mathcal{T}}} u_p \right).$$

If $\omega_{\mathcal{T}}(n) = 0$, the difference $g_X^*(n) - Z(n)$ vanishes for $|S(n)| \leq K$ and otherwise has absolute value at most $|S(n)|$. Its squared contribution is at most $\mathbb{E}_X S^4 / K^2$. If $\omega_{\mathcal{T}}(n) = 1$, with unique tail value v , the Lipschitz property of ϕ_K gives

$$|\phi_K(S + v) - S - \phi_K(v)| \leq 2|S|.$$

If $\omega_{\mathcal{T}}(n) \geq 2$, then

$$|g_X^*(n) - Z(n)| \leq |S(n)| + K(\omega_{\mathcal{T}}(n) + 1) \leq |S(n)| + \frac{3}{2}K\omega_{\mathcal{T}}(n),$$

and hence

$$|g_X^*(n) - Z(n)|^2 \leq CS(n)^2 + CK^2\omega_{\mathcal{T}}(n)(\omega_{\mathcal{T}}(n) - 1).$$

Equations (5.3)–(5.5) therefore imply

$$\mathbb{E}_X |g_X^* - Z|^2 \ll \frac{M}{K^2} + M^2 \log(2/M) + K^2 M^2 + o(1). \quad (5.6)$$

To compare g_X and g_X^* , suppose $p^k \parallel n$ with $k \geq 2$. Its contribution to the difference between the two unclipped expressions is

$$\frac{f(p^k) - f(p)}{s_X} - c_X(k - 1) \log p.$$

This tends to zero for each fixed prime power. Moreover, $\sum_{p,k \geq 2} p^{-k} < \infty$. Given $\varepsilon > 0$, choose a finite set $\mathcal{F}$ of prime powers such that $\sum_{p^k \notin \mathcal{F}, k \geq 2} p^{-k} < \varepsilon$. Then, uniformly in X ,

$$\frac{1}{X} \#\{n \leq X : \text{some } p^k \notin \mathcal{F}, k \geq 2, \text{ divides } n\} < \varepsilon.$$

Outside this exceptional set, only the fixed prime powers in $\mathcal{F}$ contribute, and their total contribution tends uniformly to zero. The Lipschitz bound and $|g_X|, |g_X^*| \leq K$ therefore give

$$\limsup_{X \rightarrow \infty} \mathbb{E}_X |g_X - g_X^*|^2 \leq 4K^2 \varepsilon.$$

Letting $\varepsilon \downarrow 0$ yields $\mathbb{E}_X |g_X - g_X^*|^2 = o(1)$; combine this with (5.6). $\square$

Lemma 5.2. *There are absolute constants $c_0, C_0 > 0$ such that*

$$\liminf_{X \rightarrow \infty} \mathbb{E}_X |z_X - \mu_{z_X}(X)|^2 \geq c_0 M - C_0 K^2 M^2. \quad (5.7)$$

Proof. Write $v_p = \phi_K(u_p)$. Since $K > 1$, (4.2) implies

$$M - o(1) \leq \sum_{p \leq X} \frac{v_p^2}{p} \leq K^2 M. \quad (5.8)$$

On $p \notin \mathcal{T}$ we have $v_p = u_p$, so (4.3) gives

$$\left| \sum_{p \leq X} \frac{v_p \log p}{p} \right| \leq |c_X| + KM \log X. \quad (5.9)$$

By partial summation from Mertens's first theorem, $\sum_{p \leq X} (\log p)^2/p \sim \frac{1}{2}(\log X)^2$. The quadratic projection identity gives

$$\begin{aligned} \inf_{b \in \mathbb{R}} \sum_{p \leq X} \frac{(v_p - b \log p)^2}{p} &= \sum_{p \leq X} \frac{v_p^2}{p} - \frac{(\sum_{p \leq X} v_p \log p/p)^2}{\sum_{p \leq X} (\log p)^2/p} \\ &\geq M - CK^2M^2 - o(1). \end{aligned} \quad (5.10)$$

Apply (2.1), dropping the nonnegative b^2 term and the terms with $k \geq 2$ from its right-hand side. Since $|v_p| \leq K$, the center estimate in Section 2 gives $|A_0(z_X, X) - \mu_{z_X}(X)| \leq K\pi(X)/X = o(1)$. The second moments are uniformly bounded by (2.3) and (5.8); changing the center consequently changes the second moment by $o(1)$. Combining this with (5.10) proves (5.7). $\square$

6 Proof of Theorem 1.1

Assume that f has no finite concentration and take g_X from (5.1). If $f(n+1) \geq f(n)$, monotonicity and Lipschitz continuity of ϕ_K give $(g_X(n) - g_X(n+1))_+ \leq |c_X| \log(1+1/n)$, where $t_+ = \max\{t, 0\}$. Using $|g_X| \leq K$ and telescoping the signed increments gives

$$\sum_{n < X} |g_X(n+1) - g_X(n)| \leq 2K + 4KD_f(X) + 2|c_X| \log X = o(X). \quad (6.1)$$

This is where clipping enters: a decrease of f may be arbitrarily large, but changes g_X by at most $2K$. For each fixed H , every difference $g_X(n) - g_X(n-j)$ is bounded by the sum of the j intervening absolute increments. Since $|g_X| \leq K$, (6.1) yields

$$\frac{1}{X} \sum_{H \leq n \leq X} |g_X(n) - \mathcal{A}_H g_X(n)|^2 \longrightarrow 0. \quad (6.2)$$

Let $m_X = \mu_{z_X}(X) - a_X$. Then $Z - m_X = z_X - \mu_{z_X}(X)$, and

$$Z - m_X = (Z - g_X) + (g_X - \mathcal{A}_H g_X) + \mathcal{A}_H(g_X - Z) + (\mathcal{A}_H Z - m_X). \quad (6.3)$$

The square of the right-hand side is at most four times the sum of the four squared terms. Jensen's inequality gives the contraction

$$\frac{1}{X} \sum_{H \leq n \leq X} |\mathcal{A}_H(g_X - Z)(n)|^2 \leq \frac{1}{X} \sum_{n \leq X} |g_X(n) - Z(n)|^2.$$

Lemma 2.1 applies to z_X , whose prime coefficients are bounded by K with quadratic mass at most K^2M . The first H indices contribute $o(1)$ by (2.3) and Cauchy-Schwarz. Taking $X \rightarrow \infty$ and then $H \rightarrow \infty$ in (6.3), and using (6.2), gives

$$\begin{aligned} \limsup_X \mathbb{E}_X |z_X - \mu_{z_X}(X)|^2 &\leq 8 \limsup_X \mathbb{E}_X |g_X - Z|^2 \\ &\leq C_1 \left(\frac{M}{K^2} + M^2 \log(2/M) + K^2 M^2 \right) \end{aligned} \quad (6.4)$$

for an absolute constant C_1 .

Choose $K > 1$ so large that $C_1/K^2 < c_0/4$, where c_0 is from Lemma 5.2. Next choose $0 < M < 1/4$ so small that

$$C_1 M \log(2/M) + (C_1 + C_0) K^2 M < c_0/4.$$

With these fixed choices, (6.4) contradicts (5.7). Hence f has finite concentration.

Erdős's Theorem V now gives (1.1). Hildebrand's Theorem 1 gives a limiting distribution for $f(n+1) - f(n)$ whose characteristic function [4, (1.4)] is a product of real factors, hence the law is symmetric. For every $\varepsilon > 0$, $\#\{n \leq X : f(n+1) - f(n) < -\varepsilon\} \leq D_f(X) = o(X)$. The limiting law assigns no mass to $(-\infty, 0)$; by symmetry it is therefore δ_0 . Hence $f(n+1) - f(n) \rightarrow 0$ in density, and a standard diagonal argument yields a density-one set along which the differences tend to zero. Hildebrand's corollary [4, pp. 258–259] now gives $f(n) = c \log n$ for some real c and all n . Finally, $c < 0$ would give a decrease at every integer, so $c \geq 0$. $\square$

Declaration of generative AI use

GPT-6 Astra was used to propose the mathematical proofs and generate the Lean formalization. GPT-5.6 Sol and Claude Opus 5 were used for editorial review of the exposition. The author finished the final manuscript and takes full responsibility for its content.

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